

James Williams

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EDUCATION

Michigan State University, College of Engineering

East Lansing, Mi

Technology Engineering, Mechatronics Concentration, CS minor, GPA: 3.5

May 2024 - Present

Mechatronics, Embedded Systems, Control Systems, Circuits, Data Structures, Machine Vision, Smart Agricultural Systems

SKILLS

- **Hardware:** PCB design, embedded circuits, wiring diagrams, breadboarding, soldering
- **Programming:** Python, C++, C#, Java, SQL, Swift
- **Robotics & Controls:** ROS2, PID tuning, GNSS/RTK, sensor fusion
- **Tools & Frameworks:** OpenCV, Git, Xcode, VSCode, CAN-bus familiarity

EXPERIENCE

Undergraduate Research Assistant – Michigan State University

East Lansing, Michigan

Department of Biosystems and Agricultural Engineering

September 2025 - Present

- Developed and validated autosteer guidance systems achieving ± 2 cm accuracy using RTK-GNSS and NTRIP corrections; performed field testing under real agricultural conditions
- Annotated and trained large datasets for AI integrations of technology
- Integrated autosteer hardware/software with tractor electrification and computer-vision weed-ID systems, ensuring cross-compatibility between autonomous subsystems
- Collaborated with farmers and engineering teams to gather user requirements and improve real-world reliability and usability of autonomous tractor platforms
- Documented experiments, field results, and design considerations in technical reports supporting ongoing research in precision agriculture automation

PROJECTS

Open-Source Contributor – AgOpenGPS Autosteer System

C#

- Contributed C# updates to an open-source autosteer platform, improving RTK-GNSS parsing reliability and tractor control responsiveness
- Implemented fixes for steering control logic, filtering, and sensor latency handling; performed bench testing and field simulation to validate behavior

Vision tracking piglet mortality reducer

- Engineered a vision-tracking pipeline using OpenCV and custom ML models to detect piglet distress behaviors in real time, improving early-warning capability for farmers
- Developed an automated behavior-monitoring system that reduces piglet mortality risk by alerting farmers to anomalies, supporting more humane and efficient livestock management
- Built and trained convolutional models using annotated datasets (600+ keypointed images) to classify piglet behavioral patterns; achieved measurable accuracy improvements across multiple training cycles

ORGANIZATIONS

Precision Agriculture Robotics Club – Autonomous Field Robot

East Lansing, Michigan

Vision & Navigation Systems Engineer

September 2024 - Present

- Built navigation pipelines in ROS2 integrating OpenCV line detection, RTK-GNSS data, and PID motor control to enable autonomous field traversal
- Trained object-detection models using RoboFlow, increasing identification accuracy to 95% for agricultural targets

- Improved system reliability through design-for-robustness practices; contributed to robotics demonstrations showcased at the ASABE national conference